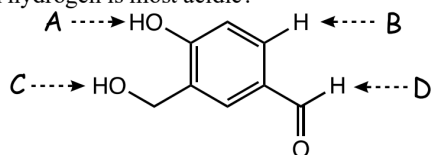
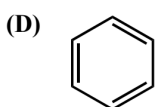
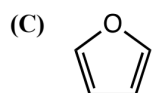


1. Which hydrogen is most acidic?

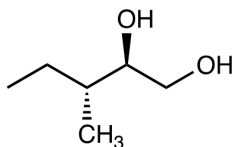
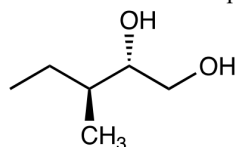


(A) A (B) B (C) C (D) D

2. In which compound are the carbon atoms not coplanar in the most stable conformation?



3. What is the relationship between the two structures?

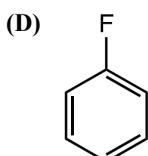
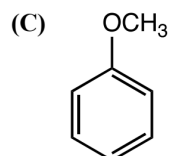
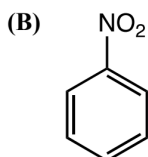
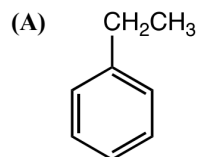


(A) Constitutional isomers (B) Diastereomers
(C) Enantiomers (D) Identical

4. What functional group is formed by the reaction of a Grignard reagent with an aldehyde?

(A) Ester (B) Ketone
(C) Secondary alcohol (D) Tertiary alcohol

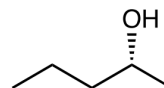
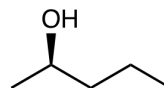
5. Which compound undergoes electrophilic aromatic substitution under the mildest conditions?



6. Which class of enzyme catalyzes the reaction of two molecules to form a larger product, usually using ATP as an energy source?

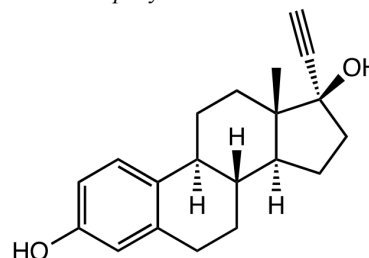
(A) Helicase (B) Ligase
(C) Protease (D) Racemase

7. What is the relationship between the two alcohols?



(A) Structural isomers (B) Enantiomers
(C) Diastereomers (D) Identical

8. In ethinylestradiol, shown below, how many sp^2 -hybridized and sp -hybridized carbons are there?



(A) 6 sp^2 , 2 sp (B) 5 sp^2 , 8 sp
(C) 11 sp^2 , 8 sp (D) 12 sp^2 , 8 sp

9. Which statement most accurately describes the most stable conformation of cyclohexane?

(A) All six carbon atoms are in the same plane.
(B) All bond angles are 109.5° .
(C) All hydrogens are eclipsed relative to the hydrogen atoms on adjoining carbons.
(D) The carbon-carbon bonds alternate between short and long distances.

10. The tertiary alcohol 2-methyl-2-propanol reacts with concentrated aqueous hydrochloric acid at 0°C to form 2-chloro-2-methylpropane. How is the mechanism of this reaction best described?

(A) S_N1 (B) S_N2 (C) E1 (D) E2

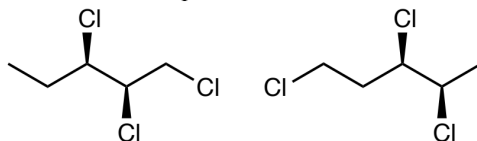
11. Which statement most accurately describes the properties of ketones and esters?

(A) Ketones are weak acids; esters are strong acids.
(B) Ketones react by nucleophilic addition; esters react by nucleophilic substitution.
(C) Ketones are readily reduced; esters are readily oxidized.
(D) The carbonyl group of a ketone is resonance stabilized; the carbonyl group of an ester is not resonance stabilized.

12. Which is not found in a typical nucleic acid?

- (A) Carboxylic acid (B) Dialkyl phosphate
(C) Heterocyclic base (D) Five-carbon sugar

13. What is the relationship between these two molecules?



- (A) Identical
(B) Enantiomers
(C) Diastereomers
(D) Structural isomers

14. Which is the strongest acid?

- (A) (B)
(C) (D)

15. Which molecule is not aromatic?

- (A) (B)
(C) (D)

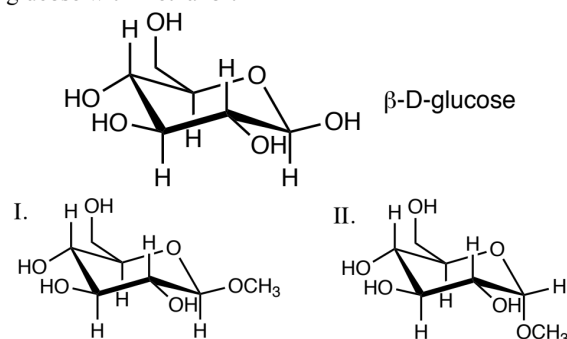
16. Which statement best describes the redox properties of alcohols?

- (A) Primary alcohols are oxidized to ketones by acidic dichromate.
(B) Ketones are reduced to primary alcohols by sodium borohydride.
(C) Secondary alcohols are oxidized to ketones by acidic dichromate.
(D) Tertiary alcohols are oxidized to esters by acidic dichromate.

17. Acetic acid reacts with ethanol in the presence of a strong acid catalyst to form ethyl acetate (the Fischer esterification reaction). What is the role of the acid catalyst?

- (A) It makes the carboxylic acid more electrophilic by protonating the carbonyl oxygen.
(B) It makes the ethanol more nucleophilic by protonating the hydroxyl oxygen.
(C) It converts the hydroxyl group of ethanol to a better leaving group.
(D) It shifts the equilibrium in the direction of the ester product.

18. Which are products of the acid-catalyzed reaction of β -D-glucose with methanol?



- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

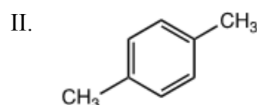
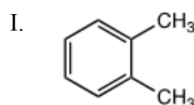
19. How many stable one-carbon molecules contain both hydrogen and chlorine and no other elements?

- (A) 2 (B) 3 (C) 4 (D) 5

20. Which property best accounts for the fact that acetic acid (CH_3COOH) is far more acidic than ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) in aqueous solution?

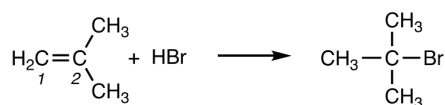
- (A) Acetic acid is more soluble in water than ethanol.
(B) Acetic acid has two hydrogen bond acceptors that can interact with water, while ethanol has only one.
(C) Acetic acid's conjugate base has its negative charge delocalized over two oxygens, while ethanol's conjugate base has its negative charge localized on one oxygen.
(D) Acetic acid can form a hydrate $\text{CH}_3\text{C}(\text{OH})_3$ in water, while ethanol cannot react with water in this way.

21. Which isomers of xylene would produce only a single mononitration product on treatment with $\text{HNO}_3/\text{H}_2\text{SO}_4$?



- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

22. Under certain conditions, HBr adds to 2-methylpropene to give predominantly 2-bromo-2-methylpropane rather than isomeric 1-bromo-2-methylpropane. Which is the best mechanistic explanation for this regioselectivity?



- (A) H^+ adds preferentially to C-1 to form the more stable carbocation.
(B) Br^- adds preferentially to C-2 to form the more stable carbanion.
(C) $\text{H}^\bullet$ adds preferentially to C-1 to form the more stable radical.
(D) $\text{Br}^\bullet$ adds preferentially to C-2 to form the more stable radical.

23. Which will most reliably distinguish between an alcohol and an ether with the formula $\text{C}_4\text{H}_{10}\text{O}$?

- (A) The alcohol will have the higher normal boiling point.
(B) The alcohol will rapidly decolorize bromine while the ether will not.
(C) The alcohol will absorb visible light while the ether will not.
(D) The alcohol will form an electrically conductive aqueous solution while the ether will not.

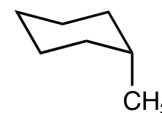
24. Which form of the amino acid alanine is most abundant in aqueous solution at $\text{pH} = 7.0$?

- (A) $\text{CH}_3\text{CH}(\text{NH}_2)(\text{COOH})$
(B) $\text{CH}_3\text{CH}(\text{NH}_2)(\text{COO}^-)$
(C) $\text{CH}_3\text{CH}(\text{NH}_3^+)(\text{COOH})$
(D) $\text{CH}_3\text{CH}(\text{NH}_3^+)(\text{COO}^-)$

25. How many stable isomers have the formula $\text{C}_2\text{H}_2\text{ClBr}$?

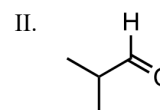
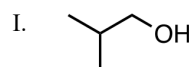
- (A) 1 (B) 2 (C) 3 (D) 4

26. Which best describes the relationship between the two structures shown below?



- (A) They are identical.
(B) They are enantiomers.
(C) They are in rapid equilibrium at room temperature and are present in equal quantities.
(D) They are in rapid equilibrium at room temperature and are present in unequal quantities.

27. Which compounds react readily with acidic dichromate solution ($\text{K}_2\text{Cr}_2\text{O}_7/\text{aqueous H}_2\text{SO}_4$)?



- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

28. In the reaction of sodium cyanide with 1-bromobutane in methanol solution, what is the principal role of the cyanide ion?

- (A) A Brønsted base
(B) A nucleophile
(C) An oxidant
(D) A solvent

29. The amino acid lysine has the side chain $\text{R} = \text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$. A mutation of lysine to which amino acid is least likely to change the overall charge on a protein at $\text{pH} = 8.0$?

- (A) Arginine, $\text{R} = \text{CH}_2\text{CH}_2\text{CH}_2\text{NHC}(=\text{NH})\text{NH}_2$
(B) Glycine, $\text{R} = \text{H}$
(C) Methionine, $\text{R} = \text{CH}_2\text{CH}_2\text{SCH}_3$
(D) Serine, $\text{R} = \text{CH}_2\text{OH}$

30. What monosaccharide units are found in both starch and cellulose?

- (A) Arabinose
(B) Fructose
(C) Glucose
(D) Ribose

31. A molecule has the formula $C_5H_{10}O$. Which functional groups might it contain?

- I. Alcohol
- II. Ketone

- (A) I only (B) II only
(C) Either I or II (D) Neither I nor II

32. How many distinct branched alkanes are there with the formula C_6H_{14} ?

- (A) 1 (B) 2 (C) 3 (D) 4

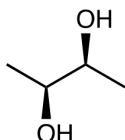
33. Which cycloalkane in its most stable conformation has average internal C–C bond angles most different from the value given?

- (A) Cyclopropane, 60°
(B) Cyclobutane, 90°
(C) Cyclopentane, 108°
(D) Cyclohexane, 120°

34. Which gas is most reactive toward 1-hexene?

- (A) N_2 (B) O_3 (C) N_2O (D) NH_3

35. Which structure depicts the enantiomer of the diol shown?



- (A)
(B)
(C)

- (D) The diol shown is achiral and does not have an enantiomer.

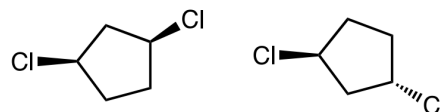
36. Which biomolecule does NOT contain C=O double bonds?

- (A) Sucrose (a carbohydrate)
(B) Glycerol trioleate (a lipid)
(C) Phenylalanine tRNA (a nucleic acid)
(D) Trypsin (a protein)

37. Which is NOT a valid representation of 3-methyl-1-butene?

- (A) $(CH_3)_2CHCH=CH_2$ (B) $CH_2CHCH(CH_3)_2$
(C) (D)

38. What is the relationship between the two compounds shown?



- (A) Identical (B) Structural isomers
(C) Geometric isomers (D) Mirror image isomers

39. An amine with the formula $C_4H_{11}N$ will have which of these properties?

- (A) Unpleasant odor
(B) Boiling point greater than $100^\circ C$
(C) Absorption of light with $\lambda > 450\text{ nm}$
(D) Water solubility less than 10 g/L

40. Which combination of reactants and catalyst will produce methyl propanoate, $CH_3CH_2COOCH_3$, upon heating?

- (A) CH_3CH_2OH and CH_3COOH with catalytic NaOH
(B) CH_3CH_2OH and CH_3COOH with catalytic H_2SO_4
(C) CH_3OH and CH_3CH_2COOH with catalytic NaOH
(D) CH_3OH and CH_3CH_2COOH with catalytic H_2SO_4

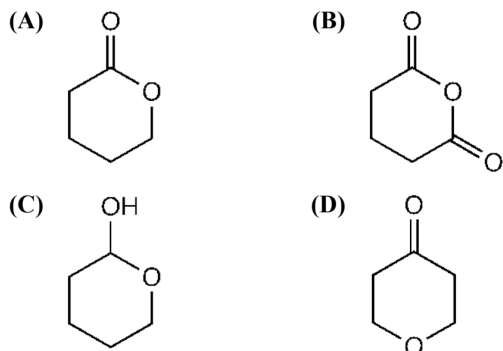
41. Which conformation of cyclohexane (C_6H_{12}) is most stable?

- (A) (B)
(C) (D)

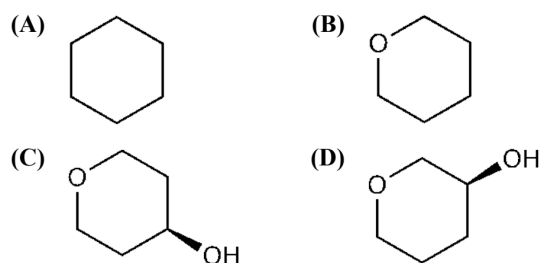
42. Which element is NOT present in DNA?

- (A) H (B) N (C) P (D) S

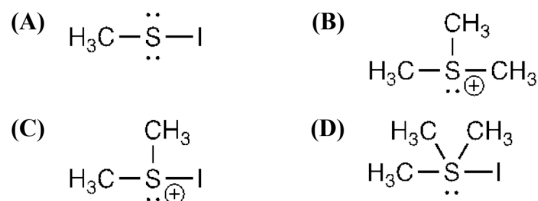
43. Which compound is an ester?



44. A pure substance is found to rotate the plane of plane-polarized light. Which compound is it?



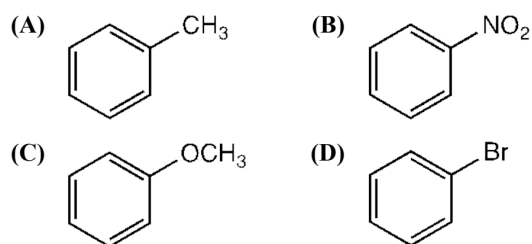
45. Which structure best represents the sulfur-containing species produced in the reaction of $(\text{CH}_3)_2\text{S}$ with CH_3I ?



46. How many distinct compounds have the formula C_5H_{12} ?

- (A) One (B) Two (C) Three (D) Four

47. Which compound reacts the fastest in electrophilic aromatic substitution reactions (such as with Br_2 in the presence of FeBr_3)?



48. Which statement about unsaturated fats is NOT correct?

- (A) Unsaturated fats contain carbon-carbon double bonds.
 (B) Unsaturated fats generally have higher melting points than saturated fats with the same number of carbon atoms.
 (C) Unsaturated fats can be converted to saturated fats by treatment with H_2 in the presence of a metal catalyst.
 (D) Unsaturated fats must contain oxygen.

49. Which compound can exhibit geometric isomerism?

- (A) 1-butene (B) 2-butene
 (C) 1-butyne (D) 2-butyne

50. Which structural features contribute to the low strain energy of cyclohexane?

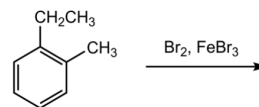
- I. All bond angles are close to 109.5° .
 II. All C-H bonds on adjacent carbon atoms are staggered.

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

51. What reaction conditions most effectively convert a carboxylic acid to a methyl ester?

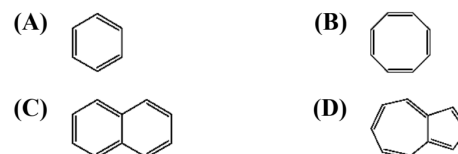
- (A) CH_3OH , HCl (B) CH_3I , HCl
 (C) CH_3OH , NaOH (D) CH_3I , SOCl_2

52. What is the maximum number of monosubstitution products of the aromatic substitution reaction shown?



- (A) 1 (B) 2 (C) 3 (D) 4

53. Which compound is NOT considered aromatic?



54. To what elements are the phosphorus atoms in DNA bonded?

- I. C
II. H

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

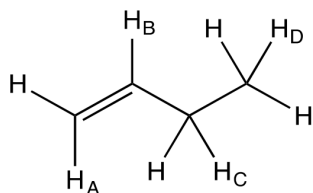
55. How many π bonds are in a molecule of propyne, C_3H_4 ?

- (A) 0 (B) 1 (C) 2 (D) 3

56. How many isomers have the formula $C_2H_2Br_2$?

- (A) 1 (B) 2 (C) 3 (D) 4

57. One hydrogen in 1-butene is replaced by bromine to give a chiral molecule. Which hydrogen is replaced?

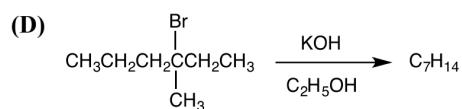
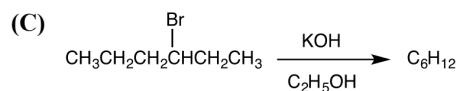
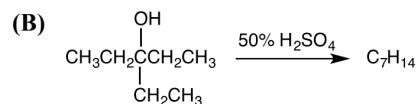
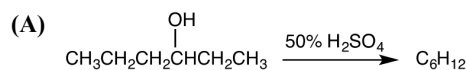


- (A) H_A (B) H_B (C) H_C (D) H_D

58. A compound with the formula C_6H_{12} does not decolorize bromine in $CHCl_3$ solution. Which compound could this be?

- (A) Cyclohexane (B) 1-Hexene
(C) *Trans*-3-hexene (D) *Cis*-3-hexene

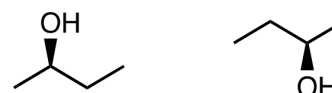
59. Which reaction would yield a single alkene product?



60. The net chemical reaction of photosynthesis is best described as

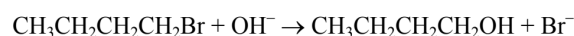
- (A) an endothermic reaction that forms sugars.
(B) an exothermic reaction that forms sugars.
(C) an endothermic reaction that breaks down sugars.
(D) an exothermic reaction that breaks down sugars.

61. What is the relationship between the following two molecules?



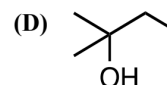
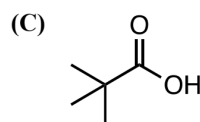
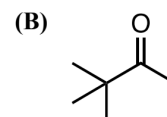
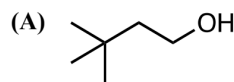
- (A) Structural isomers (B) Geometric isomers
(C) Enantiomers (D) Identical

62. What is the role of hydroxide ion in the reaction below?

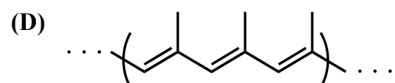
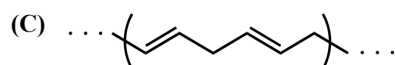
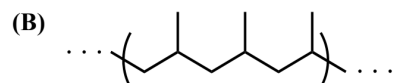
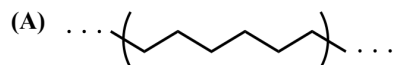


- (A) Oxidizing agent (B) Lewis acid
(C) Catalyst (D) Nucleophile

63. Which compound reacts readily with $Cr(VI)$ reagents?



64. What is the structure of polypropylene, formed by the polymerization of propene ($CH_2=CHCH_3$)?



Organic/Biochemistry

65. The sequence of amino acids in a protein is known as its
(A) primary structure. (B) secondary structure.
(C) tertiary structure. (D) quaternary structure.
66. Molecules from which class of biopolymers can react with water, in the presence of suitable enzymes, to form smaller examples of that class of biopolymers?
I. Proteins II. Polysaccharides
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
67. How many sigma bonds are in 2-butyne ($\text{CH}_3\text{C}\equiv\text{CCH}_3$)?
(A) 3 (B) 6 (C) 9 (D) 11
68. Which alcohol will undergo acid-catalyzed dehydration under the mildest conditions?
(A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ (B) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
(C) $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$ (D) $(\text{CH}_3)_3\text{COH}$
69. A 1.0 M aqueous solution of which compound has the lowest pH?
(A) $\text{CH}_3\text{CH}_2\text{OH}$ (B) CH_3COOH
(C) CH_3CHO (D) Cl_3CCHO
70. A six-carbon organic compound containing oxygen is suspected of being either a secondary alcohol or a ketone. Which chemical or physical test would best distinguish between these two possibilities?
(A) Water solubility
(B) Melting point
(C) Treatment with sodium bicarbonate
(D) Treatment with acidic dichromate
71. Which two bases are found as a hydrogen-bonded base pair in DNA?
(A) A and T (B) C and T
(C) C and U (D) G and U
72. What is the nature of the peptide bonds in a protein?
(A) Hydrogen bonds (B) Amide bonds